

UNIT 1

MATERIALS WE USE

Learning objectives:

- Learn about natural and man-made materials and their uses.
- Understand the properties of materials (hardness, strength, flexibility).
- Explore how materials are chosen based on their properties.
- Identify materials in daily life through a Material Hunt activity.
- Reflect on the importance of material properties in product selection.

Everything around us is made from different materials. Materials are substances used to make objects. Some are natural, like wood and cotton, while others are man-made, like plastic and metal.

1.1 Types of Materials

1. Natural Materials: These are materials that come from nature. They are found in the earth, plants, or animals. People do not make them.

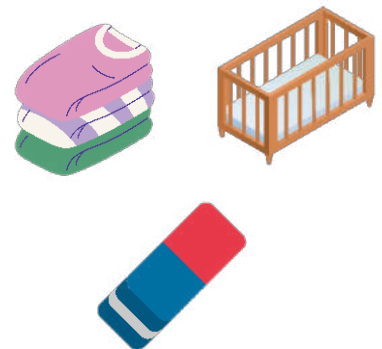
•**Examples:** Wood from trees, cotton from plants, wool from sheep, rocks, and water.

Uses of Natural Materials:

Wood: It is used for making furniture and paper.

Cotton: It is used for making clothes.

Rubber: It is used for making tires and erasers.



2. Man-Made Materials: These are materials made by people using natural materials or chemicals. They are changed or created in factories.

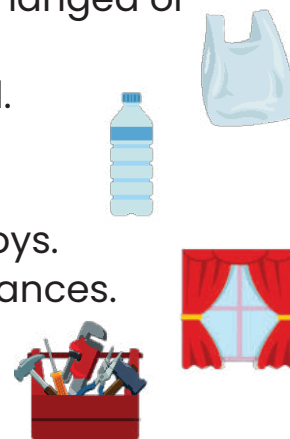
•**Examples:** Plastic, glass, paper, bricks, and steel.

Uses of Man-Made Materials:

Plastic: It is used for making bottles, bags, and toys.

Metal: It is used for making cars, tools, and appliances.

Glass: It is used for making windows, mirrors, and jars.



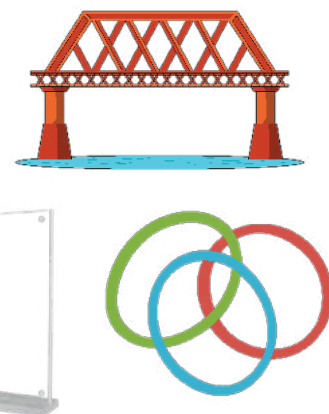
1.2 Properties of Materials

Different materials have different properties:

Hard or Soft: Some materials are hard, like metal, while others are soft, like cotton.

Strong or Weak: Some materials can hold heavy things without breaking, like metal, while others cannot.

Flexible or Rigid: Some materials bend easily, like rubber, while others do not, like glass.



1.3 Uses of Materials in Daily Life

We use different materials to make everything we see around us i.e. clothes, shoes, toys, books, furniture, and more. Understanding the different types of materials helps us choose the right ones for different purposes.



Fill in the blanks.

1. Wood, cotton, and rubber are examples of _____ materials.
2. _____ is a man-made material used for making bottles and toys.
3. _____ is a natural material used for making clothes.
4. Some materials are _____ and can bend easily, like rubber.
5. _____ is a hard material often used for making cars and tools.



Tick (✓) for True and cross (x) for False statements.

1. Cotton is a man-made material. ()

2. Metal is a strong material that can hold ()

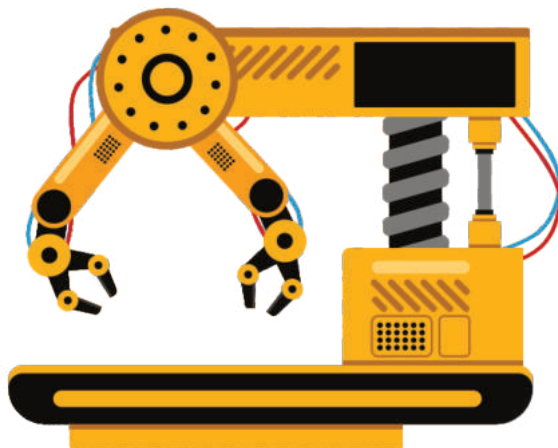
heavy things.

3. Glass is a flexible material that bends easily. ()

4. Rubber is used to make erasers and tires. ()

5. We use different materials to make clothes, ()

furniture, and toys.



Multiple Choice Questions.

1. Which of these is a natural material?

- a) Plastic
- b) Metal
- c) Wood
- d) Glass

2. What is plastic commonly used to make?

- a) Clothes
- b) Bottles and toys
- c) Tyres
- d) Mirrors

3. Which material is known for being flexible?

- a) Metal
- b) Glass
- c) Rubber
- d) Wood

4. Which property best describes cotton?

- a) Hard
- b) Soft
- c) Rigid
- d) Transparent

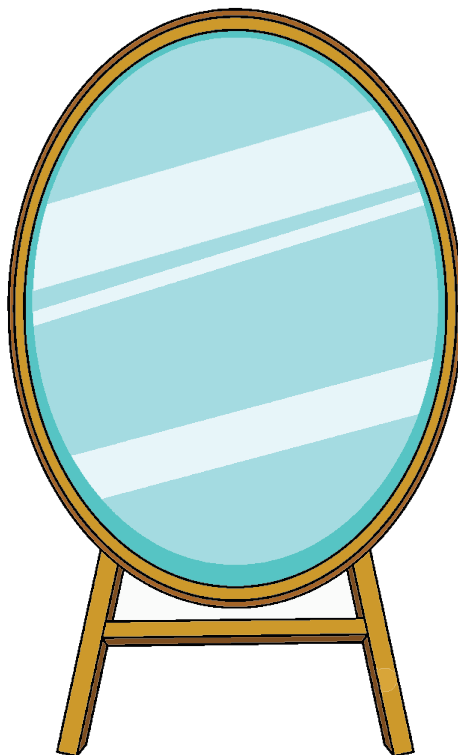


5. What material would you use to make a mirror?

- a) Metal
- b) Wood
- c) Glass
- d) Rubber

6. Why do we use different materials for different purposes?

- a) Because all materials are the same
- b) Because materials have different properties
- c) Because materials are only natural
- d) Because materials do not break



Test Your Understanding

Q1: What is the difference between natural materials and man-made materials? Give two examples of each.

Ans: _____

Q2: What is wood commonly used for?

Ans: _____

Q3: Describe the difference between flexible and rigid materials?

Ans: _____

Q4: Why is it important to know the properties of different materials?

Ans: _____



Q5: Give an example of a strong material and explain why it is strong?

Ans: _____



Critical Thinking

Can you name two materials that come from nature and two that are made by humans? Why are both types of materials important in our daily lives?





Material Hunt



ACTIVITY

Identify different materials and their uses in your surroundings.

Materials Needed

Notebook

Pencil

Instructions:

1. Walk around your home or classroom and find objects made from different materials (wood, metal, plastic, glass, etc.).
2. Write down each object and the material it is made of.
3. Describe why you think that material was chosen to make the object. For example, "A spoon is made of metal because it is strong and doesn't break easily."



ACTIVITY

What are they made of?

